

Optimization of Energy-Efficient Cloud Radio Access Networks for 5G using Neural Networks

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Abstract—Since proposed, Cloud Radio Access Network (Cloud-RAN) gives a committed architecture suitable for fulfilling 5G networks' applications. Cloud-RAN can solve challenges related to ever-evolving networks' mobile operators and an ever-growing number of end-users. Cloud-RAN architecture maintains both profitability and quality of service (QoS). In this paper, power consumption is jointly formulated as power minimization beamforming and RRHs selection problem. Using the conventional convex or heuristic optimization approaches to find optimal solutions is highly complex; hence, we introduce an Artificial Neural Network (ANN) - based optimization model that aims to optimize the active Remote Radio Heads (RRHs) numbers in remote network sites and the consumed power. The proposed model considers various signal to interference plus noise ratios per client and power consumption models. Specifically, the model uses an adopted Bi-Section Group Sparse Beamforming (GSBF) optimization algorithm to reach near optimum solutions. Obtained validated results encourage machine learning techniques to reduce both the complexity and power consumption in such an emerging area.

Keywords— Artificial Neural Network (ANN), Cloud Radio Access Network (Cloud-RAN), Group Sparse Beamforming, Remote Radio Heads (RRHs) optimization

I. INTRODUCTION

Recently, Mobile data traffic has increased exponentially due to a growing number of smart devices and their applications in the network. By 2023, Cisco expects that up to 71% of the global population will have mobile connectivity, and the total number of mobile subscribers will grow to 5.7 billion [1]. As a result, mobile network operators have to improve the coverage and capacity of existing radio access networks to catch this exponential growth in data demand [2] and maintain a satisfying level of quality of service (QoS) for end-users. Towards facing these challenges, many potential solutions have been discussed. For example, building and operating new RANS or even upgrading the old ones and applying the small-cell networks [3]. Although these solutions have increased network performance in coverage and capacity and decreased network losses, they have unwanted effects on the network power consumption [2]. The total cost of ownership (TCO), is increased; meanwhile, the Average Revenue per User (ARPU) remains stable or even falls over time. Tidal effect pattern for the mobile network, resulting in oversubscribed base stations (BSs) in some areas and wasted

power to others [2], mobile operators search for new network architecture capable of increasing the performance and reducing those challenges. Cloud-RAN comes as an efficient solution.

Cloud-RAN architecture details in [2][4]. It consists of three main parts: i) Pool of baseband Units (BBUs): performing baseband processing tasks and base station functions. ii) Remote Radio Heads (RRHs): responsible for radio functions at remote sites and iii) Fronthaul network: high bandwidth and low-latency connection between BBU pool and RRHs. Cloud-RAN is a centralized-green network architecture. It saves up to 80% of the power in large networks compared to traditional ones. Centralized management and operation impacts in reducing network expenses.. Furthermore, sharing active users' signaling, traffic data, and Channel State Information; impacts capacity improvement [2]. As well, it enables a non-uniform loading adaptation. Despite the significant reduction in power consumption that Cloud-RAN architecture offers, many RRHs are deployed at the remote sites carrying data. Thus, front-haul links power consumption and RRH selection in Cloud-RAN architecture cannot be ignored and should be studied [5].

Recently, many works focused on the Base station or RRH selection problem in the Cloud-RAN system, which has a noticeable effect on network performance, particularly power consumption [6]. For example, in [7], the transmit precoding matrices and active RRH set are jointly optimized, targeting minimizing the consumed power in the network for a user-centric Cloud-RAN. The formulated optimization problem is solved through two stages of low complexity proposed algorithms. Additionally, a Markov decision process (MDP) based model for optimal RRH selection in Cloud-RAN was proposed in [8], which serves end users connection demands by minimizing the blocking probability and maximize revenue.

Although solving the above models via conventional optimization methods leads to optimal results, but it still has a high complexity; thus, the machine learning are widely used due to their great capacities in dealing with such complex models [9]. In the same context, researchers presented the importance of incorporating ANN models in wireless networks to cope with design complexity and fulfil 5G requirements [10]. ANNs are a subfield of artificial intelligence that replicate biological neural networks' structure [11], consists of artificial neurons connected with input, output, and hidden layers in an organized format.

For detailed recent works of ML models for Cloud-RAN are fully surveyed in [12]. Specifically, in [13], the design of the RRH selection and transmit beamforming problem was optimized to achieve the minimum consumed power. The joint problem was modelled using a fully connected feed-forward neural network composed of four layers; input, output, and hidden layers with ReLU activation function. This network accepts inputs as channel vectors and outputs as beamforming vectors of active RRHs. Similarly, in [14], a deep neural network was constructed to determine the set of optimal active RRHs and then obtain corresponding beamforming via solving a designed convex optimization problem. The proposed approach targets minimizing the power consumption; fulfilling users demands.

Motivated by the above advantages of ML models, this paper proposes a comprehensive work that combines artificial intelligence with Cloud-RAN. An ANN is trained and designed to optimize and achieve an optimum set of active RRHs at remote sites. Accordingly, optimal power consumption and minimal optimization problem complexity are achieved.

A. Contribution

This paper aims to achieve a green Cloud-RAN system that optimizes consumed power at cell site RRHs and in the connection transport network; satisfies QoS needs of end-users. The optimization problem was formulated as a joint RRH selection and power minimization beamforming[15]. Network power consumption is determined as the number of running RRHs at remote sites, and transmitted power of such active RRHs is minimized through a coordinated beamforming approach. In [16], a comprehensive study aimed at increasing Cloud-RAN performance was presented. There, the heuristic driven approach was applied to obtain near optimum solutions. Results were encouraging; however, the algorithm suffered high complexity, especially in large scale Cloud-RAN. Improving such work may need machine learning to overcome complex recursive optimization techniques. Therefore, an ANN-based optimization model is proposed to select an efficient number of RRHs input to the power optimization algorithm. Then the trained model is applied to the proposed Bi-Section GSBF optimization algorithm to surpass the heuristic driven approach [16] in terms of total consumed power and time complexity.

B. Organization

The remainder of this paper is structured as follows. Section 2 presents the Cloud-RAN system, power consumption models, and related reformulation analysis. Section 3 discusses the proposed optimization model based on neural networks. Section 4 will discuss the simulation results. Finally, Section 5 summarizes the paper and discusses potential future work.

II. CLOUD-RAN SYSTEM MODEL

A. Architecture model

The presented system architecture is illustrated in Fig.1 [15][16]. It is composed of L deployed RRHs of the same type, each connected with N antennas, and K end users, each with a single antenna: $S = \{1, \dots, K\}$ denotes users' index set. RRHs are deployed and distributed remotely and connected to a centralized BBU pool through fiber links modelled as passive optical network (PON) for energy and cost-saving. Baseband

processing functions are shifted to the BBU, where end users' data and CSI are available. This enables resource sharing and cooperation between RRHs; the more active and cooperative RRHs, the total transmission power is reduced. Links power in Cloud-RAN system has a significant impact on increasing the overall network power consumption. Thus, minimizing the number of active RRHs and their corresponding links will improve Cloud-RAN performance related to consumed power.

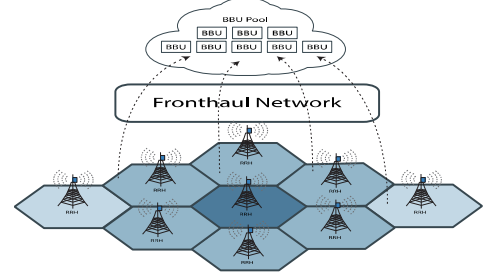


Fig. 1. The architecture of the system model [15].

Paper Notation: $(\cdot)^T, (\cdot)^H, \Re(\cdot), \mathbb{C}$, and $\mathbb{E}[\cdot]$ denotes the transpose, Hermitian, and fundamental part and complex domains, the expectation of a random variable, respectively. $\mathcal{CN}(0, \sigma_k^2)$ denotes a complex Gaussian distribution. $|\cdot|$ stands for either set size or real scalar absolute value, depending on the context, while $\|\cdot\|_p$ denotes ℓ_p -vector norm. Assume $\mathcal{L} = \{1, \dots, L\}$ the index set for RRHs. A and Z denote the active and the inactive RRH set, respectively, where $A \subseteq \mathcal{L}$ & $A \cup Z = \mathcal{L}$.

At Remote Site, a baseband signal X_l is transmitted from each RRH, calculated as:

$$X_l = \sum_{k=1}^K W_{lk} S_k, \forall l \in A, \quad (1)$$

where $W_{lk} \in \mathbb{C}^{N_l \times 1} \forall l \in \mathcal{L}, \forall k \in S$ is the beamforming coefficients at RRH l for end-user k , S_k is user k data symbol with unit power $\mathbb{E}[|S_k|^2] = 1$. Transmitted power for each active RRH is constrained by:

$$\sum_{k=1}^K \|W_{lk}\|_{l_2}^2 \leq P_l, \forall l \in A, \quad (2)$$

where P_l is the maximum power transmitted from RRH l .

At the End User, a baseband signal y_k is received at user k .

$y_k = \sum_{l \in A} h_{kl}^H W_{lk} S_k + \sum_{i \neq k} \sum_{l \in A} h_{kl}^H W_{li} S_i + E_k$, (3) where $h_{kl} \in \mathbb{C}^{N_l \times 1} \forall l \in \mathcal{L}, \forall k \in S$ is the channel vector between user k and RRH l , E_k is the additive noise with Gaussian distribution $\mathcal{CN}(0, \sigma_k^2)$, and σ_k^2 is the noise power at end user k . In the case of a low interference area where interference can be considered as noise, each user has a signal-to-interference-plus-noise ratio (SINR) of:

$$\text{SINR}_k = \frac{|\sum_{l \in A} h_{kl}^H W_{lk}|^2}{\sum_{i \neq k} |\sum_{l \in A} h_{kl}^H W_{li}|^2 + \sigma_k^2} \forall l \in \mathcal{L} \quad \forall k \in S. \quad (4)$$

B. Power consumption model

A sleep mode for RRHs and connected links is considered:

1) Model of RRH power

Using the linear model in [17] for RRH power consumption:

$$P_l^{\text{rrh}} = \begin{cases} P_{a,l}^{\text{rrh}} + \frac{1}{\eta_l} P_l^{\text{out}} & \text{if } P_l^{\text{out}} > 0, \\ P_{s,l}^{\text{rrh}}, & \text{if } P_l^{\text{out}} = 0, \end{cases} \quad (5)$$

where $P_{a,l}^{rrh}$ is the active consumed power for RRH l , $P_{s,l}^{rrh}$ is the sleep consumed power for RRH l , P_l^{out} is the transmitted power for RRH l , and η_l is antenna efficiency RRH l power amplifier. This linear model of RRHs considers the power of the whole RRH system, and it did not calculate the subsystem components of each RRHs. This fits well with the proposed optimization problem goals, as we need only RRHs power consumption at on / off states. Other models are proposed to capture the power consumed by each sub-component of the RRH. In [18], the RRH power model is broken down into five sections: Number of antennas, RF Transceiver, power amplifier (PA), RRH's power conversion, RRH cooling.

2) Fronthaul links the power model

Considering PON front-haul network [19], where it's Optical Network Units (ONUs) implemented in a sleep mode, but the Optical Line Terminal (OLT) is always on [20]. Thus, power consumed in the front-haul network is defined as:

$$P^{tn} = P_{olt} + \sum_{l=1}^L P_l^{tl}, \quad (6)$$

where P_{olt} is the OLT consumed power, $P_l^{tl} = P_{a,l}^{tl}$, and $P_l^{tl} = P_{s,l}^{tl}$ denote the power consumed by the ONU l in the active and the sleep modes, respectively.

3) Overall network consumed power

As described in [15], $P_l^a \triangleq P_{a,l}^{rrh} + P_{a,l}^{tl}$ ($P_l^s \triangleq P_{s,l}^{rrh} + P_{s,l}^{tl}$) is defined as the active (sleep) power consumption when both the RRH and connected link are in on (off) case. Therefore, the overall power consumption of the network is given by:

$$\begin{aligned} \hat{P}(A) &= \sum_{l \in A} \frac{1}{\eta_l} P_l^{out} + \sum_{l \in A} P_l^a + \sum_{l \in Z} P_l^s + P_{olt} \\ &= \sum_{l \in A} \frac{1}{\eta_l} P_l^{out} + \sum_{l \in A} P_l^a \\ &\quad + \sum_{l \in L} P_l^s - \sum_{l \in A} P_l^s + P_{olt} \\ &= \sum_{l \in A} \frac{1}{\eta_l} P_l^{out} + \sum_{l \in A} (P_l^a - P_l^s) \\ &\quad + \sum_{l \in L} P_l^s + P_{olt} \\ &= \sum_{l \in A} \sum_{k=1}^K \frac{1}{\eta_l} \|W_{lk}\|_{l_2}^2 + \sum_{l \in A} P_l^c + \underbrace{\sum_{l \in L} P_l^s + P_{olt}}_{\text{Const term with given } L} \end{aligned} \quad (7)$$

where P_l^c is the link consumed power for simplification. By considering the constant term as in "(7)," the total power is:

$$P(A, W) = \underbrace{\sum_{l \in A} \sum_{k=1}^K \frac{1}{\eta_l} \|W_{lk}\|_{l_2}^2}_{\text{Total transmit power}} + \underbrace{\sum_{l \in A} P_l^c}_{\text{Total relative transport link power}}. \quad (8)$$

C. Problem formulation and analysis

The problem of minimizing network power consumption can be formulated in this work as a combination of RRH set selection and beamforming transmission.

$$\mathcal{P} : \underset{\{W_{lk}\}, A}{\text{minimize}} \quad P(A, W)$$

$$\text{Subject to} \quad \frac{|\sum_{l \in A} h_{kl}^H W_{lk}|^2}{\sum_{i \neq k} |\sum_{l \in A} h_{kl}^H W_{li}|^2 + \sigma_k^2} \geq \gamma_k, \quad \sum_{k=1}^K \|W_{lk}\|_{l_2}^2 \leq P_l, l \in A, \quad (9)$$

where γ_k is user k target SINR.

This joint problem should be reformulated so that beamforming phases do not affect the objective function or constraints. As a result, the SINR constraint can be reformulated to a second-order cone (SOC) constraint [21] as:

$$C_Q(A, W) : \left\| \frac{[h_k^H W_1, \dots, h_k^H W_K]^T}{\sigma_k} \right\| \leq \sqrt{\beta_k} \Re(h_k^H W_k), \quad (10)$$

where $\beta_k = 1 + \frac{1}{\gamma_k}$.

Also, the RRH power constraints in SOC form is defined as:

$$C_p(A, W) : \|\tilde{W}_l\| \leq \sqrt{P_l}, l \in A, \quad (11)$$

where $\tilde{W}_l = [W_{l1}^T, \dots, W_{lK}^T]^T \in \mathbb{C}^{KN_l \times 1}$

The resulting problem with the predefined SOC constraints is NP-hard with convex cardinality is hard to solve in polynomial time. Therefore, the cardinality (A) should be fixed, and the problem becomes convex-SOCP [21]. Modelling frameworks (e.g., CVX) [22] can be used to find the solution. As a result, the driven optimization problem with the given active RRH set is:

$$\begin{aligned} \mathcal{P}(A) : \underset{W}{\text{minimize}} \quad & \sum_{l \in A} \sum_{k=1}^K \frac{1}{\eta_l} \|W_{lk}\|_{l_2}^2 \\ \text{Subject to} \quad & C_Q(A, W), C_p(A, W), k \in S \end{aligned} \quad (12)$$

D. ANN Architecture and performance analysis

We designed an artificial neural network model with feed-forward topology, as shown in Fig.2, which has one input layer, one output layer and two hidden layers (i.e. each with 15 neurons). A linear transfer function is used in the output layer, and hyperbolic tangent transfer functions are used in hidden layers for non-mapping operations and adjusting the neuron's outputs. It accepts three inputs sets; maximum transmit power consumption for RRHs (p_l set), target SINRs for end-users (γ_k set), and relative power consumption to connected front-haul links (p_l^c set); where: p_l set = $[p_1, p_2, \dots, p_L] \in \mathbb{R}^{1 \times L}$ watt, γ_k set = $[\gamma_1, \gamma_2, \dots, \gamma_K] \in \mathbb{R}^{1 \times K}$ dB, and p_l^c set = $[p_1^c, p_2^c, \dots, p_L^c] \in \mathbb{R}^{1 \times L}$ watt. The output is the optimum number of active RRHs $|A^*|$. The proposed ANN is trained using supervised learning (i.e. Bayesian regularization process), where the number of instances is equal to 250 (80% train and 20% test, randomly selected). These instances are collected from simulation. The simple structure of this model is well suited to the small scale structure of the network model mentioned in this paper and the simulation results. An extensive scale network may require other advanced models of ANN to deal with a large number of parameters affecting network decisions.

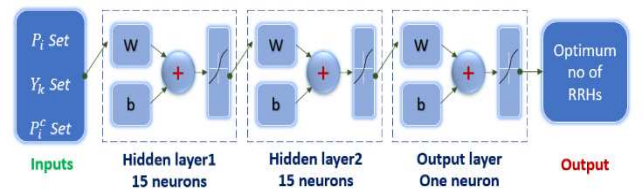


Fig. 2. ANN design model for optimum RRH-Selection.

Three different criteria measure the performance of this model: Mean Square Error (MSE), Regression, and Error Histogram between network outputs and desired targets. Fig.3 shows MSE errors versus no epochs, where the model starts with a significant MSE value then decreases for both training and testing sets until it flattens off at an acceptable level with a small MSE value. Regression analysis for the proposed network model is measured, as shown in Fig.4, which indicates the model yields a training performance (correlation coefficient R) of 0.99996, testing performance 0.99044. The overall correlation coefficient of the network was 0.99754 showing the best fitting and strong correlation between outputs and corresponding targets in the network and the highest degree of reliability and accuracy to predict new instances.

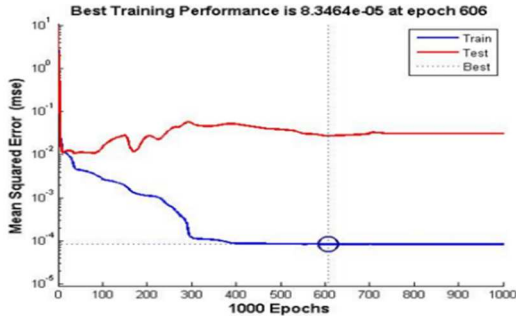


Fig. 3. Mean squared error for trained and tested data versus no of epochs.

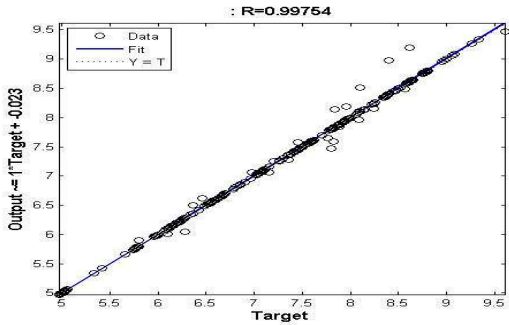


Fig. 4. Regression for output data versus target to trained and tested sets.

Error Histogram analysis with 20 bins is plotted for both trained and tested data sets in Fig.5. As seen, most instances have errors near zero, implying the accuracy and correctness of most of the model outputs, and the error is near zero, which proves the accuracy and correctness of the proposed model.

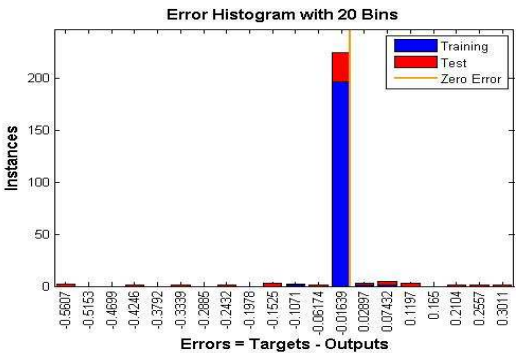


Fig. 5. Error Histogram for both trained and tested sets.

III. ARTIFICIAL NEURAL NETWORK-BASED OPTIMIZATION MODEL FOR BI-SECTION GSBF ALGORITHM

In Fig.6, the proposed trained ANN model for optimum RRH selection to Bi-Section GSBF algorithm is illustrated. First, the algorithm checks the network state (i.e., number of distributed RRHs, end-users & QoS requirements and front-haul power consumption). Then, the proposed ANN model delivers the near-optimum number of RRHs needed to be active in the remote site. After that, the Bi-Section GSBF algorithm is applied to reach optimum and optimized solutions in some running RRHs and consumed power.

A. Bi-Section GSBF algorithm procedure

Bi-Section GSBF algorithm proposed in [15] is a binary search algorithm to minimize the number of active RRHs and hence total power consumption by applying three stages GSBF framework: **Stage 1**, it minimizes a weighted mixed (ℓ_1/ℓ_2 -norm) group-sparsity inducing norm problem defined in "(14)," cause the group sparsity in the aggregative beamforming w .

$$w = [\underbrace{w_{11}^T, \dots, w_{1K}^T}_{\tilde{w}_1^T}, \dots, \underbrace{w_{L1}^T, \dots, w_{LK}^T}_{\tilde{w}_L^T}]^T, \quad (13)$$

where $\tilde{w}_l = [w_{l1}^T, \dots, w_{lK}^T]^T \in \mathbb{C}^{KN_l}$, form a group.

$$\begin{aligned} \mathcal{P}_{GSBF} : \underset{w}{\text{minimize}} \quad & 2 \sum_{l=1}^L \sqrt{\frac{P_l^c}{\eta_l}} \|\tilde{w}_l\|_{l_2} \\ \text{Subject to} \quad & C_Q(L) \quad C_p(L). \end{aligned} \quad (14)$$

As a result of the previously defined weights for the beamformers coefficients groups, RRHs with a higher link power consumption and a lower power amplifier efficiency will have a greater chance of being forced to be turned off. $\mathcal{P}_{GSBF}$ solved as SOCP problem. **Stage 2**, Based on some key system parameters: beamforming gain, channel power gain, power amplifier efficiency, and relative link power consumption, an ordering rule is applied to set the priority for the off RRHs.

$$\theta_l := \sqrt{\frac{k_l \eta_l}{P_l^c}} (\sum_{k=1}^K \|\hat{w}_{lk}\|_{l_2}^2)^{1/2}, \quad \forall l = 1, \dots, L, \quad (15)$$

where the RRH with a lower θ_l had a higher priority for deactivation. **Stage 3** is a binary search procedure (i.e., the Bi-Section search method) to determine the optimal active RRH set, followed by coordinated beamforming.

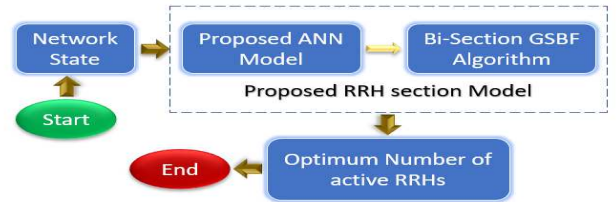


Fig. 6. A proposed ANN model based Bisection GSBF Block diagram.

B. The proposed ANN model complexity analysis

The proposed trained ANN model is applied to the Bi-Section GSBF algorithm for optimum RRH selection, the complexity of optimization problems, and the total number of iterations. As a result, the total number of SOCP problems that must be solved grows quadratically with a near-optimal solution of the active number of RRHs obtained by the ANN model (L_{ANN}) rather than the total number of presented RRHs (L),

given that the overall time complexity of the Bi-Section GSBF algorithm is given by [15][23]:

$$O((N_l L K)^{3.5} \text{Log } L), \quad (16)$$

where L is the total number of RRHs at the site, N_l is the number of antennas at RRH l and K is the total number of users. By applying the proposed model, the complexity is:

$$O((N_l L_{ANN} K)^{3.5} \text{Log } L_{ANN}), \quad (17)$$

where L_{ANN} is near the optimum solution of the active number of RRHs obtained by the ANN proposed model. Since $L_{ANN} < L$, we achieve a reduction in time complexity.

TABLE I. SIMULATION PARAMETERS

Parameter	Value
Standard deviation log-norm shadowing σ_s	8dB
Path-loss at distance d_{kl} (km)	$148.1 + 37.6 \log_2(d_{kl})$
Small-scale fading distribution $\dot{g}_{kl}$	$\mathcal{CN}(0, 1)$
Noise power σ_k^2 (10MHz bandwidth)	-102 dBm
Maximum transmit power of RRH P_l	1 W
Transmit antenna power gain	9dBi
Power amplifier efficiency η_l [17]	25%
Confidence Level	95%
Max number of epochs	1000

IV. SIMULATIONS AND RESULTS

Some simulation results are analyzed quantitatively, the deployed network's performance improvement can be achieved concerning running RRHs at the remote site, and the average total consumed power. Furthermore, we assess the time complexity of the Bi-Section GSBF algorithm adopting optimal solutions for designing a power-efficient system. Obtained results are compared with Coordinated Beamforming (CB) technique [5], which only optimize the total transmit power, considering all deployed RRHs, and links are active. A channel modelled between the user k and the active RRH l as in [15].

$$h_{kl} = 10^{-\frac{L(d_{kl})}{20}} \sqrt{\varphi_{kl} \delta_{kl} \dot{g}_{kl}}, \quad (18)$$

where $L(d_{kl})$ is the calculated path-loss at a distance d_{kl} , φ_{kl} is the antenna gain, δ_{kl} is the shadowing coefficient and $\dot{g}_{kl}$ is the small-scale fading coefficient. Table I shows the standard cellular network settings used [15]. The obtained results have been statistically confirmed with a 95% confidence interval. For the power model described in "(7)", we set $P_{s,l}^{rrh} = 4.3 \text{ W}$ and $P_{s,l}^{tl} = 0.7 \text{ W}$, $\forall l$, and $P_{olt} = 20 \text{ W}$ [17][20]. The adaptive algorithms are defined in simulation results as follows: **"Bi-Section GSBF"**: represents Bi-Section GSBF algorithm [15]. **"Near-optimum Bi-Section GSBF"**: the near-optimum solution to Bi-section GSBF algorithm [16]. **"Proposed ANN model"**: near-optimum solution obtained by a proposed ANN. All simulations results are done using MATLAB (i.e., ML Toolbox+ CVX +SDPT3 solver [22]).

A. Power consumption versus target SINR for end-users

Consider a network with ten distributed RRHs, each with 2-antennas and 15 mobile users, each with single-antenna distributed uniformly and independently in a square region "2000 m $\times$ 2000 m," for comparison with results in [15][16]. All the transport link power are set to be $P_l^c = (5 + l) \text{ W}$, $l = 1, \dots, L$. Averaging across 50 network realizations generated randomly for simulation points. The proposed ANN model has

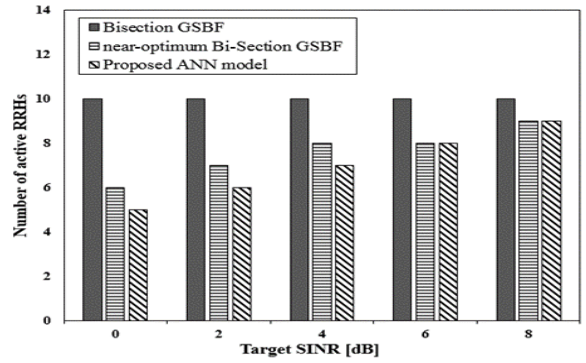
reached a better near an optimum number RRHs at different SINRs, surpassing both near-optimum Bi-Section GSBF and Bi-Section GSBF, as shown in Fig.7. The optimized number of active RRHs at remote sites driven by the proposed ANN model impacts the total average of network and relative power of active frontal links. Fig.8 demonstrates the average network power consumption at target SINRs; shows a reduction in average network consumed power, especially at low SINRs of clients. The 95 % confidence interval is used to validate these results.

Obtaining the near-optimum results is a critical point from a time complexity point of view. The ANN model-based needs approximately 2.5 minutes compared to about 24 minutes in the heuristic approach, reflecting the ANN model's efficiency. Table II shows a clear comparison between the three approaches after starting the Bisection GSBF algorithm with near outputs from the ANN model. The heuristic approach and starting algorithm with all network RRHs are active, which shows a time improvement by applying the proposed model.

TABLE II. ALGORITHMS EXECUTIVE TIME COMPARISON (MINUTES)

Approach	Bi-section GSBF	Near-optimum	Proposed ANN
Executive time	24.0383	19.68	11.0733

Fig. 7. Improvement in active RRHs versus target SINR when using the



proposed ANN corresponding to existing models.

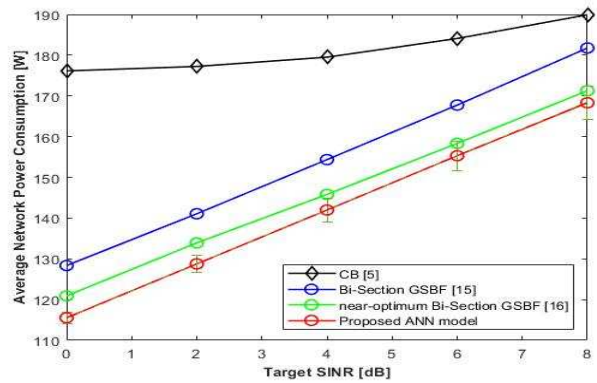


Fig. 8. Average network power consumption versus links power when using the proposed ANN corresponding to existing models.

B. Overall network power versus links power

In the same previous network deployment, by setting all links power equal, and the threshold SINR for all users is 4 dB Fig.9 shows the estimated number of RRHs at different values of relative power consumption and indicates that the ANN

model predict the optimum active RRHs, especially at high values of relative power. Fig. 10 illustrates the average network power consumption versus relative transport link power. This figure shows a minimized power consumption by applying the proposed ANN model as a pre-stage to the Bi-Section GSBF algorithm compared to a heuristic-driven approach.

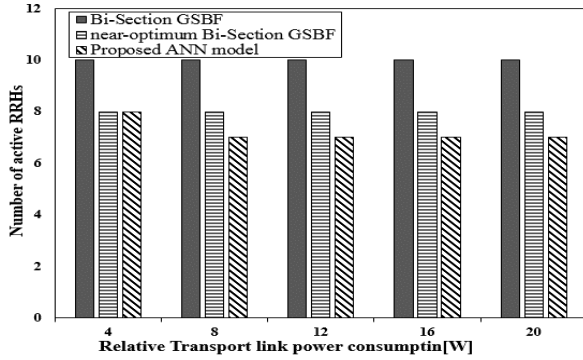


Fig. 9. Improvement in active RRHs versus links power when using the proposed ANN corresponds to existing models.

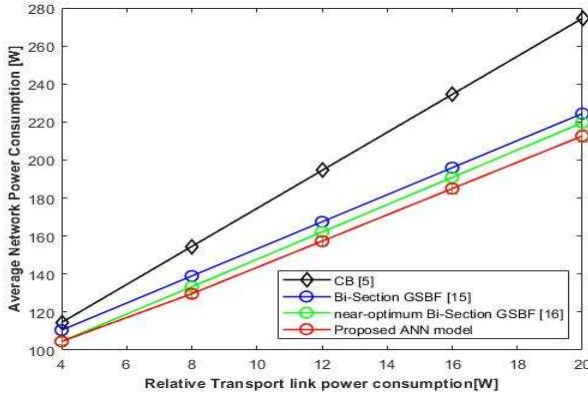


Fig. 10. Average network power consumption versus links power when using the proposed ANN corresponding to existing models.

V. CONCLUSION

In this work, improving the performance of the Cloud-RAN system through minimizing network power consumption, optimizing RRH selection, and their corresponding connected links were studied. An ANN-based optimization model for optimized RRH selection was presented. The model was tested through the Bi-Section GSBF algorithm to reach near-optimum solutions of consumed power. Simulation results demonstrated better-optimized solutions. The active number of RRHs has been reduced by around 10% at low SINRs compared to the near optimum. Moreover, the time complexity for obtaining optimum active RRHs numbers and overall simulation time was reduced compared to an employed heuristic-driven approach. The total complexity reduced by about 10% and more than 60% in the whole simulation process. Since our model is a supervised learning approach-based optimization algorithm, thus as a future work, we will investigate other ML techniques such as deep learning that can learn the feature patterns and predicts active RRHs and BS beamforming simultaneously.

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